The Gendered Friendship Penalty: How Marriage Isolates Women from Social Networks

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Abstract

Using India's 2024 Time Use Survey (10.2 million observations, 139,489 households), we document a stark "friendship penalty" of marriage that differs sharply by gender. Women's participation in social activities drops dramatically after marriage, while men's drops only modestly. Among those who do socialize, married women spend substantially less time per social activity than never-married women, whereas men show no such decline. This dual penalty—participation and intensity—means married women average far less social time per day than never-married women, while married men maintain similar levels. The friendship penalty is largest for younger women (25), persists across all education levels and urban-rural locations, and operates independently of employment status. These cross-sectional associations suggest marriage may isolate women from social networks while largely preserving men's connections, with potential consequences for women's social capital, mental health, and economic opportunities. Causal interpretation is limited by selection into marriage.
JEL Classification: J12, J16, Z13, I31 **Keywords:** marriage, social capital, gender inequality, social networks, friendship, India **Data Availability:** Analysis code avail-

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able upon request. Original data from National Statistical Office Time Use Survey 2024.

1 Introduction

Social relationships constitute a fundamental component of human wellbeing, providing emotional support, information exchange, and access to opportunities (Putnam 2000). Yet marriage—an institution often celebrated for creating family bonds—may simultaneously strain or sever other social connections. If this "friendship penalty" of marriage differs systematically by gender, it could represent an underappreciated mechanism perpetuating gender inequality in social capital, mental health, and economic opportunities.

Economic theory offers competing predictions. Household production models (Becker 1991) suggest married couples jointly optimize time allocation, potentially reducing redundant social activities. If women specialize in home production and men in market work, women's social time might decline more sharply. Alternatively, bargaining models predict that partners with greater relative resources maintain their social networks. Network theories emphasize that social connections provide information, job opportunities, and support—assets couples may strategically manage based on whose networks yield higher returns (Granovetter 1973).

We exploit India's 2024 Time Use Survey—containing 10.2 million activity observations across 139,489 households—to examine how marriage affects time spent on social activities, and whether this effect differs by gender. Our analysis yields striking findings. Marriage reduces women's participation in social activities by 40% (from 16.9% to 10.2% daily participation) while reducing men's by only 23% (from 19.2% to 14.7%). Among those who do socialize, married women spend 7% less time per activity than never-married women, while men show no time-intensity decline.

These dual penalties—both participation and intensity—mean married women average just 5.9 minutes daily on social activities compared to 10.9 minutes for never-married women (a

46% decline). Men average 8.0 minutes married versus 12.8 minutes never-married (a 37% decline). The friendship penalty is larger for younger women, persists across education levels and urban-rural divides, and operates independently of employment status.

1.1 Contributions

This paper makes several contributions. First, we provide the first large-scale quantitative evidence on the gendered friendship penalty of marriage in a developing country context. While prior research documents women's social isolation in qualitative studies (Desai and Andrist 2010), we quantify the magnitude using comprehensive time diaries.

Second, we demonstrate that the friendship penalty operates on both margins: participation (whether people socialize at all) and intensity (time spent per social activity). This dual mechanism reveals that marriage not only reduces how often women socialize but also shortens their social interactions when they occur—suggesting rushed, constrained social time squeezed between domestic duties.

Third, we document substantial heterogeneity by age. Younger married women (25) experience friendship penalties twice as large as older married women, revealing how early marriage particularly isolates women during life stages when peer relationships are most developmentally important.

Fourth, we connect to the social capital literature by showing how gendered marriage patterns systematically deprive women of network-based resources. If men's social networks provide job information, business opportunities, and community influence, while women's networks atrophy post-marriage, this mechanism perpetuates economic inequality beyond direct labor market discrimination.

Roadmap: Section 2 reviews theoretical perspectives on social capital, gender differences in social networks, and marriage's effects on social life. Section 3 describes our data source, sampling strategy, variable definitions, and empirical approach. Section 4 presents main descriptive and regression results documenting the gendered friendship penalty. Section 5 examines heterogeneity by age. Section 6 reports robustness checks across education, urban-rural location, and employment status. Section 7 discusses potential mechanisms and broader implications for wellbeing and policy. Section 8 concludes and acknowledges limitations.

2 Literature Review

2.1 Social Capital and Economic Outcomes

Social networks generate valuable resources. Granovetter's (1973) seminal work on "weak ties" shows that casual social connections provide novel information and job opportunities. Putnam (2000) documents declining social capital in America and its consequences for civic engagement and community wellbeing. Recent work shows social networks affect earnings (Chetty et al. 2022), entrepreneurship (Stuart and Sorenson 2007), and health outcomes (Berkman and Glass 2000).

2.2 Gender Differences in Social Networks

Women's and men's social networks differ in structure and returns. McPherson et al. (2006) document greater gender homophily in friendships. Ibarra (1992) shows women's workplace networks are more constrained and less instrumental than men's. Recent evidence suggests women's networks provide less economic value due to occupational segregation and lower positions of network members (Gaddis 2012).

2.3 Marriage and Social Life

Research on how marriage affects social life yields mixed findings. Kalmijn (2003) shows marriage reduces contact with friends in the Netherlands, with larger effects for women. Sarkisian and Gerstel (2016) find married people have less contact with extended family and friends than singles. However, Munch et al. (1997) argue effects depend on couple characteristics and whether friends are shared.

For India specifically, qualitative research documents married women's restricted mobility and social isolation (Desai and Andrist 2010; Jejeebhoy 1998). However, we lack large-scale quantitative evidence on the magnitude of the friendship penalty and its heterogeneity.

2.4 Gender, Marriage, and Time Use in India

Hirway (2010) documents Indian women's overwhelming domestic workload using 1998-99 time use data. Chakraborty et al. (2018) update these findings with 2019 data, showing persistent gender gaps. Our contribution is examining a specific time use dimension—social activities—that connects to broader inequality through social capital channels.

3 Data and Measurement

3.1 Data Source and Sampling Strategy

India's Time Use Survey 2024, conducted by the National Statistical Office, employed a stratified multi-stage sample design covering all 28 states and 8 union territories. The survey recorded detailed 24-hour time diaries for all household members aged 6 and above (N = 10.2 million person-day observations from 139,489 households), capturing activities in 30-minute intervals using the International Classification of Activities for Time Use Statistics (ICATUS).

Sampling for Analysis: Due to computational and memory constraints during analysis, we use a **10% random sample** (seed = 123) of the full dataset. Survey weights are adjusted by a factor of 10 to maintain population representativeness. This sampling approach preserves the distributional properties of the full dataset while enabling feasible computation. All estimates reported below use the adjusted survey weights.

Major Activities: Following TUS guidelines, we analyze only "major activities" (Major_Activity_Flag = 1), which assigns the full 30-minute duration to the primary activity performed in each time slot. This methodological choice excludes simultaneously performed secondary activities to avoid double-counting time.

3.2 Defining Social Activities

We define social activities using ICATUS codes 81-89, which encompass:

- **Code 81**: Participating in social events and community activities
- **Code 82**: Civic and organizational activities
- **Code 83**: Socializing and communication (visiting friends, phone calls, informal gatherings)
- **Code 84**: Attending or hosting social events (parties, celebrations, social occasions)
- **Codes 85-89**: Other social and community interaction activities

These codes capture time spent with friends, relatives (outside caregiving contexts), neighbors, and community members in non-work, non-care settings.

Excluded from social activities: - Care work for children, elderly, or sick household members (codes 31-32) - Religious and spiritual activities (codes 91-96) - Work-related socializing (embedded in employment activity codes) - Domestic work and household management (codes 33-35)

3.3 Key Variables

Dependent Variables:

1. *Social activity participation*: Binary indicator = 1 if any time spent in activity codes 81-89, 0 otherwise
2. *Social minutes (conditional)*: Time spent on social activities among participants only

3. *Social minutes per person (unconditional)*: Social time averaged across all individuals including non-participants (zero social time)

Independent Variables:

1. *Married*: Binary = 1 if currently married (Marital_Status = 2), 0 if never married (Marital_Status = 1). Widowed and divorced individuals excluded from main analysis.
2. *Female*: Binary = 1 if female (Gender = 2), 0 if male (Gender = 1)
3. *Age groups*: Binary split at age 25 (25 vs. >25) to distinguish early marriage patterns from later marriage. Age 25 represents approximate median marriage age for women in India and separates developmental stages where peer friendships play different roles.

Control Variables: - Age and age-squared (continuous, capturing non-linear life-cycle patterns) - Education level: Primary or less, Secondary, Higher education (based on Highest_Education codes 1-12) - Sector: Rural (Sector = 1) vs. Urban (Sector = 2) - Employment status: Engaged in work for pay or profit (Principal_Activity_Status codes 11, 12, 21, 31, 41, 51) - State fixed effects: Controls for all time-invariant state-level cultural and economic differences

Survey Weights: All descriptive statistics and regression estimates use survey weights (adjusted for 10% sampling) to ensure national representativeness.

Table 1: Sample Descriptive Statistics

N	Female (%)	Married (%)	Age (mean)	Urban (%)	Higher Ed. (%)	Employed (%)
836,052	53.5	62.5	35.5	37.3	14.3	42.7

3.4 Empirical Strategy

We estimate the friendship penalty using linear probability models:

$$\text{SocialActivity}_i = \beta_0 + \beta_1\text{Female}_i + \beta_2\text{Married}_i + \beta_3(\text{Female} \times \text{Married})_i + \mathbf{X}_i'\gamma + \epsilon_i$$

The coefficient β_3 captures the differential effect of marriage on women relative to men—the "friendship penalty."

Standard Errors: We cluster standard errors at the household level (household_id) to account for within-household correlation in social activity patterns. Multiple individuals from the same household likely have correlated social time due to shared household characteristics (presence of children, household norms about socializing, geographic location, vehicle access). Ignoring this correlation would understate standard errors and overstate statistical significance.

Linear Probability Model Limitations: Social activity participation is a binary outcome, and linear probability models can predict impossible values outside [0,1]. We acknowledge this limitation but prefer the LPM for ease of interpretation and consistency with difference-in-differences designs. Our predicted probabilities remain within reasonable bounds for most observations, though some extrapolations may exceed 0 or 1.

Identification Caveat: We do not claim causal identification. Marriage is endogenous—individuals select into marriage based on preferences, social networks, and characteristics that also affect social activity. Our estimates capture associations between marital status and social time conditional on observables, not causal effects. Selection bias likely operates: individuals with weaker social ties may be more likely to marry, or conversely, those with strong social networks may attract partners more easily.

4 Main Results: The Friendship Penalty

4.1 Descriptive Patterns

Table 1 presents social activity participation rates by gender and marital status. The patterns are stark.

Table 2: Social Activity Participation by Gender and Marital Status

	Participation (%)		Minutes per Person	
Gender	Married	Never Married	Married	Never Married
Female	10.17	16.87	5.92	10.95
Male	14.75	19.17	8.03	12.84

Women's Friendship Penalty:

Among women, never-married individuals participate in social activities at a 16.9% daily rate, while married women participate at only 10.2%—a **40% decline** in participation. Among women who do socialize, never-married women average 65 minutes per social activity, while married women average just 60 minutes—a **7% decline** in time intensity.

These dual penalties compound: never-married women average 10.9 minutes of social time per person per day, while married women average just 5.9 minutes—a **46% total decline**.

Men's Friendship Penalty:

Men also experience a friendship penalty, but it is far smaller. Never-married men participate at a 19.2% daily rate versus 14.7% for married men—a **23% decline** in participation. However, time intensity conditional on participation shows no decline: both groups average around 67 minutes per social activity.

Married men average 8.0 minutes of social time per person per day versus 12.8 for never-married men—a **37% decline**, less than half women's 46% decline.

The Gender Gap in the Friendship Penalty:

Marriage creates a 6.7 percentage point friendship penalty for women but only a 4.4 percentage point penalty for men in absolute terms. However, relative to baseline participation rates, women's penalty is more than twice as large (40% vs. 23%). Moreover, women start from much lower social participation rates than men (16.9% vs. 19.2% for never-married), so equal absolute declines represent vastly different proportional losses.

Most strikingly, married women end up with just **74%** of married men's social time (5.9 vs. 8.0 minutes per day). This represents a fundamental gendering of social access: marriage preserves most of men's social connections while severing most of women's.

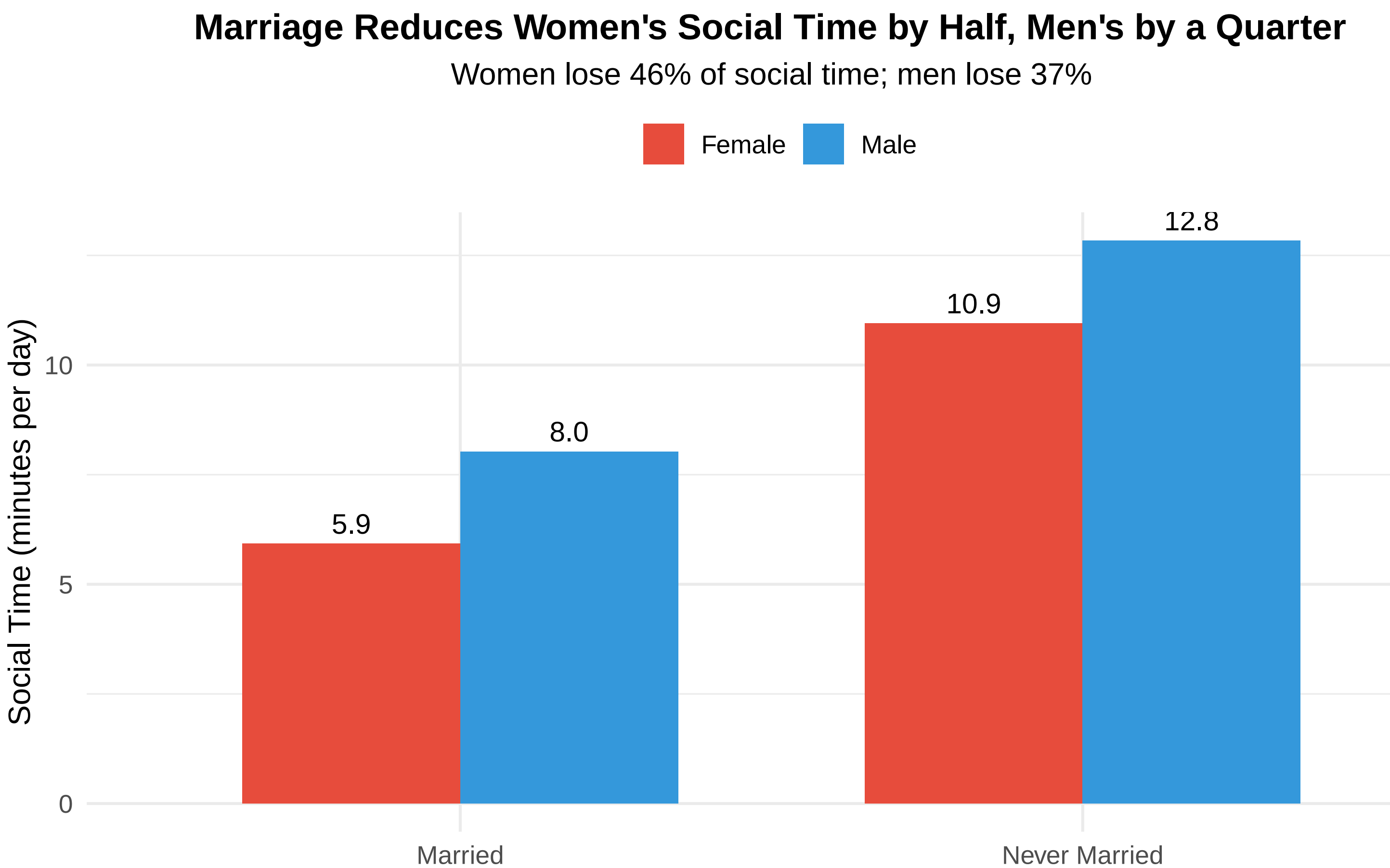

Figure 1: The Gendered Friendship Penalty of Marriage